

Task Specification
(Start, Target, End, Map)

Layer 1: Distance calculation
between coordinate points via the
proposed improved A* algorithm

Construction of cost matrix
 $D[N \times N]$

Layer 2: Open Traveling Salesman
Problem (OTSP) optimization via
MMAS-VND-CL

Visiting sequence
 π^*

Path planning via the proposed
improved A* algorithm

